Experiment No. 6 (Software) **Date: 06.12.2022**

Shift Registers

Aim:

To design shift registers using Verilog programming.

Software Required:

ModelSim

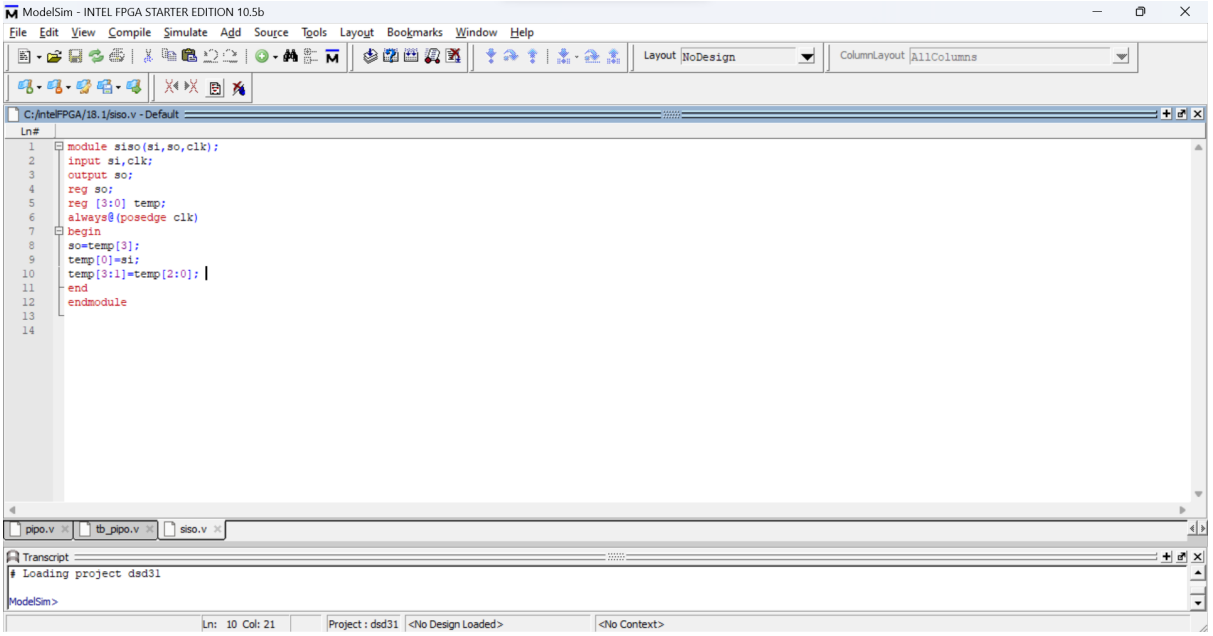
Procedure:

1. Open ModelSim.
2. Create New Project, then create a new file by selecting the verilog in the drop-down.
3. Type the code, then compile it.
4. Type the test bench, compile and simulate it.
5. Click add wave in the window and then run it.

Program Code:

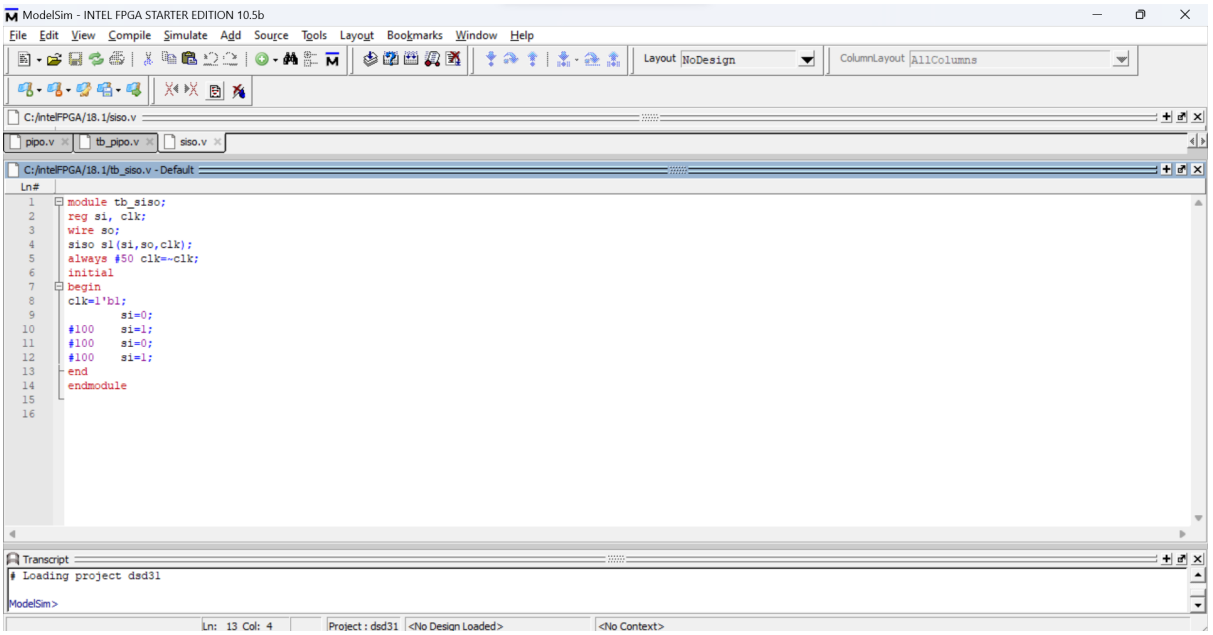
SISO:

```
module siso(si,so,clk);
input si,clk;
output so;
reg so;
reg [3:0] temp;
always@(posedge clk)
begin
so=temp[3];
temp[0]=si;
temp[3:1]=temp[2:0];
end
endmodule
```



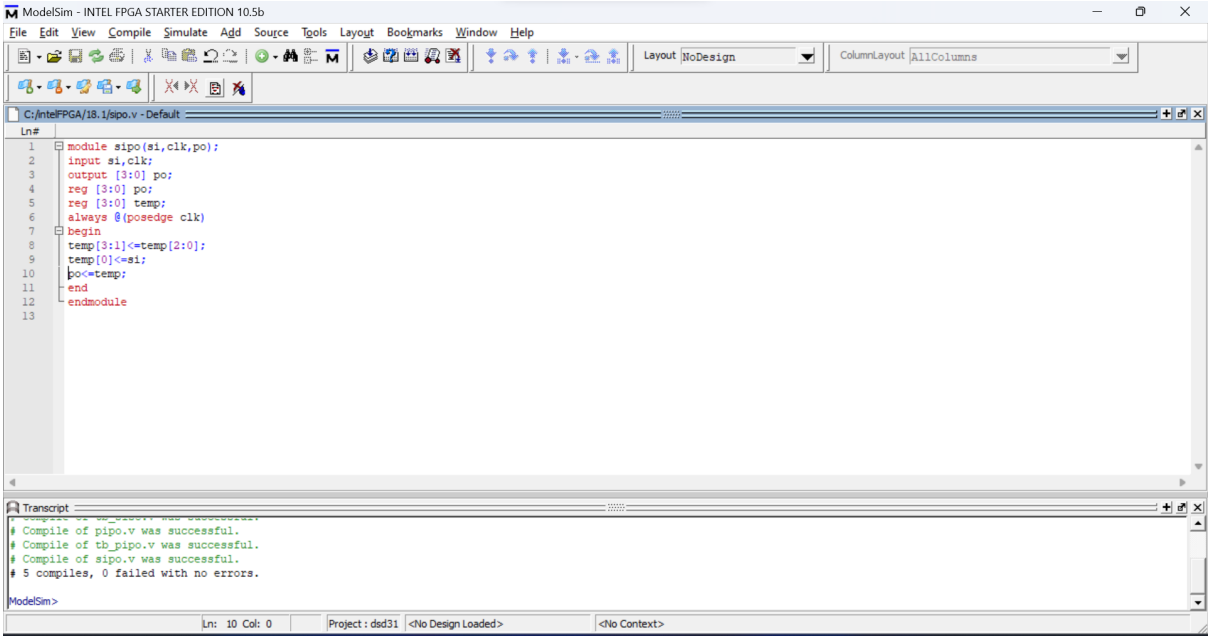
Test Bench for SISO:

```
module tb_siso;
reg si, clk;
wire so;
siso s1(si,so,clk);
always #50 clk=~clk;
initial
begin
clk=1'b1;
        si=0;
#100    si=1;
#100    si=0;
#100    si=1;
end
endmodule
```



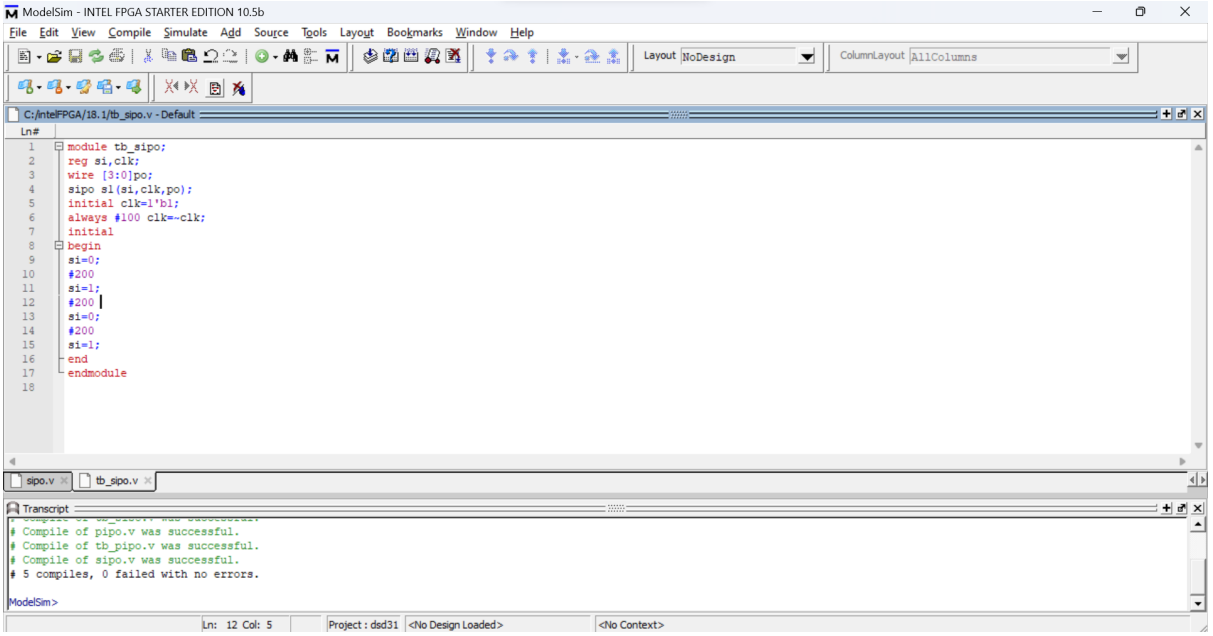
SIPO:

```
module sipo(si,clk,po);
input si,clk;
output [3:0] po;
reg [3:0] po;
reg [3:0] temp;
always @(posedge clk)
begin
temp[3:1]<=temp[2:0];
temp[0]<=si;
po<=temp;
end
endmodule
```



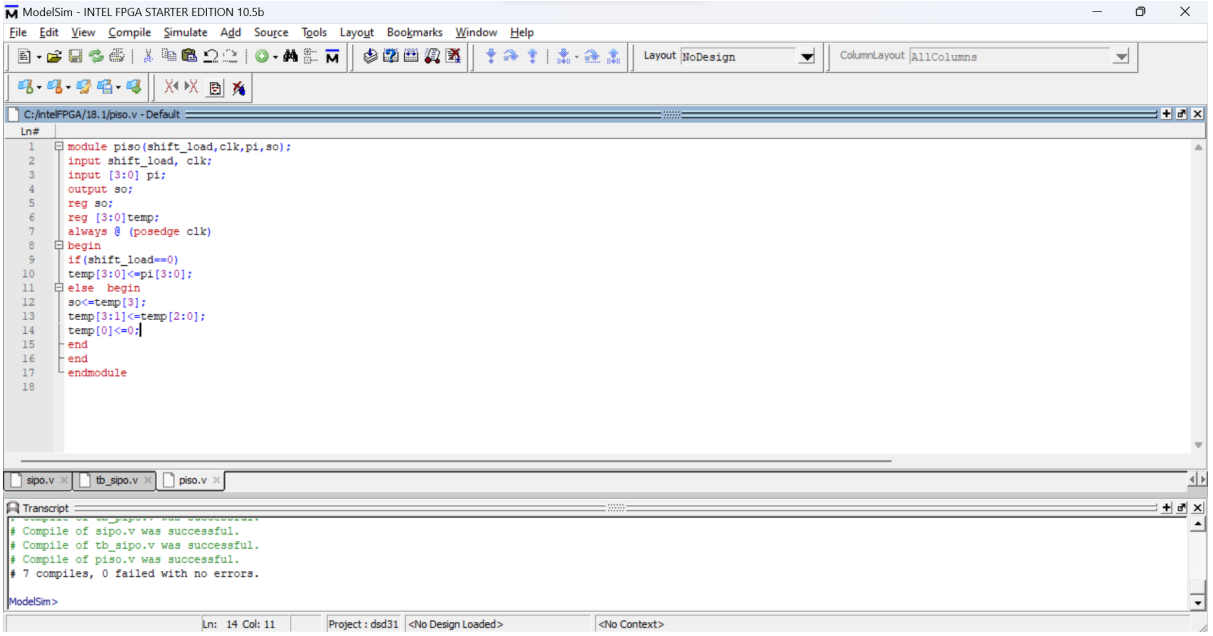
Test Bench for SIPO:

```
module tb_sipo;
reg si,clk;
wire [3:0]po;
sipo s1(si,clk,po);
initial clk=1'b1;
always #100 clk=~clk;
initial
begin
si=0;
#200
si=1;
#200
si=0;
#200
si=1;
end
endmodule
```



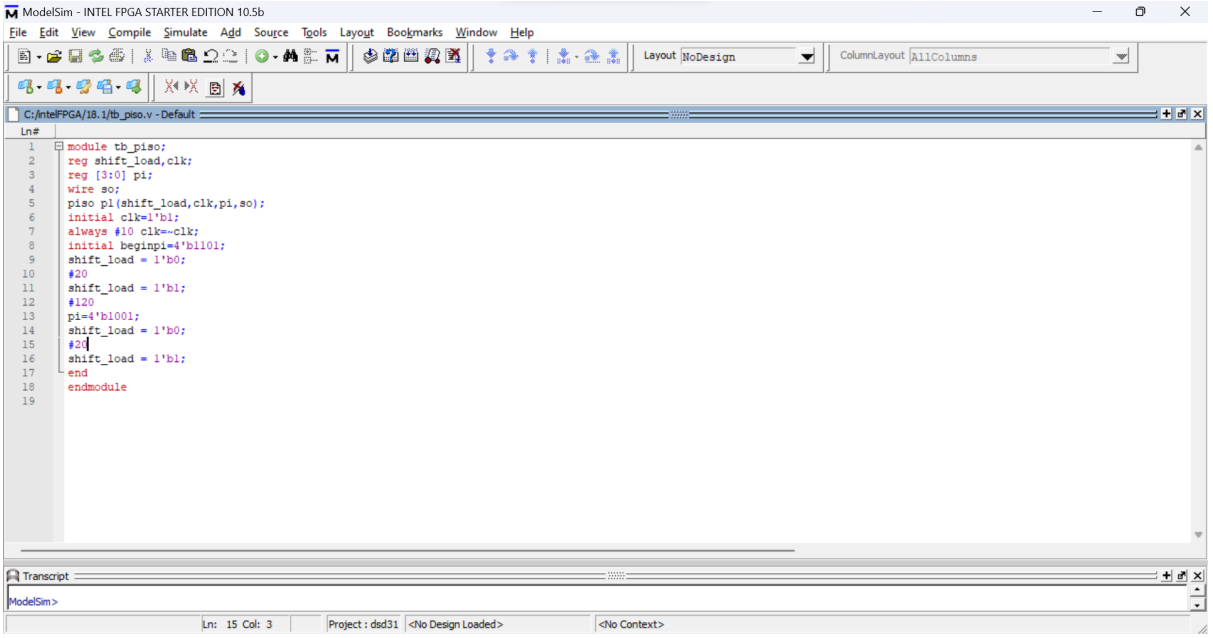
PISO:

```
module piso(shift_load,clk,pi,so);
input shift_load, clk;
input [3:0] pi;
output so;
reg so;
reg [3:0]temp;
always @ (posedge clk)
begin
if(shift_load==0)
temp[3:0]<=pi[3:0];
else  begin
so<=temp[3];
temp[3:1]<=temp[2:0];
temp[0]<=0;
end
end
endmodule
```



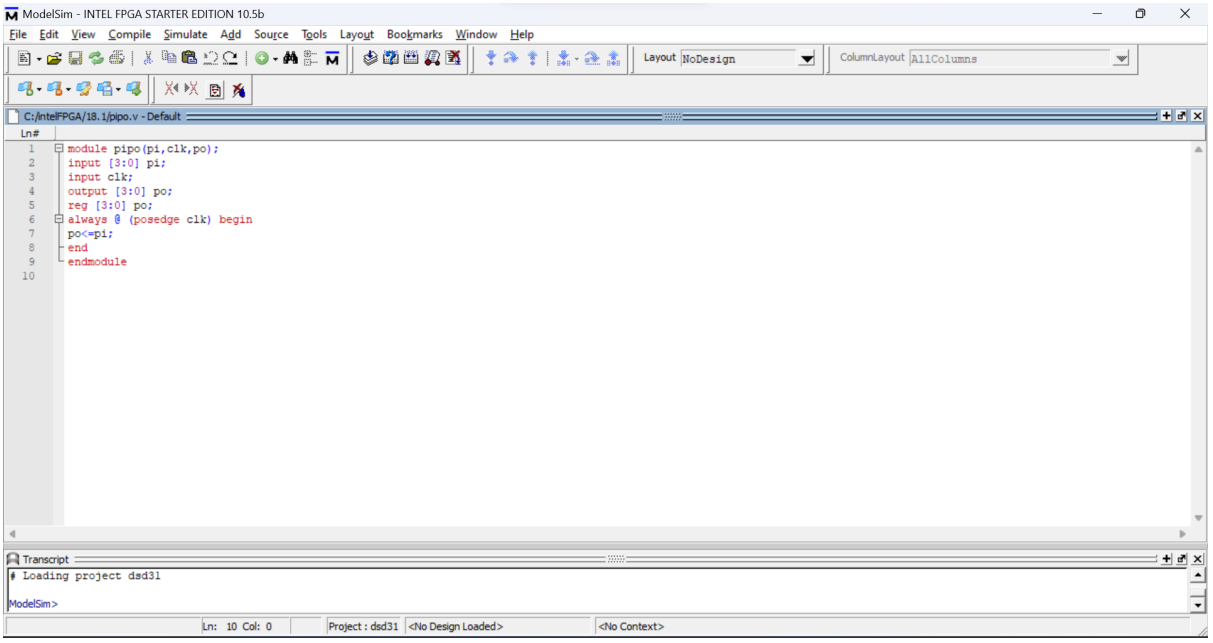
Test Bench for PISO:

```
module tb_piso;
reg shift_load,clk;
reg [3:0] pi;
wire so;
piso p1(shift_load,clk,pi,so);
initial clk=1'b1;
always #10 clk=~clk;
initial beginpi=4'b1101;
shift_load = 1'b0;
#20
shift_load = 1'b1;
#120
pi=4'b1001;
shift_load = 1'b0;
#20
shift_load = 1'b1;
end
endmodule
```



PIPO:

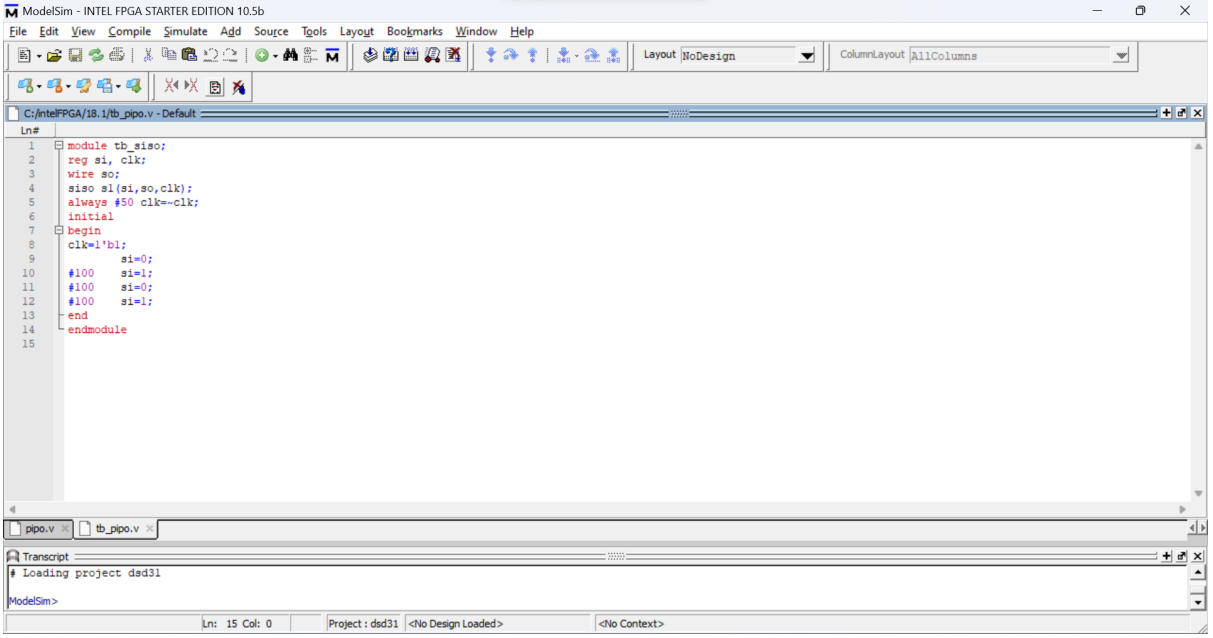
```
module pipo(pi,clk,po);
input [3:0] pi;
input clk;
output [3:0] po;
reg [3:0] po;
always @ (posedge clk) begin
po<=pi;
end
endmodule
```



Test Bench for PIPO:

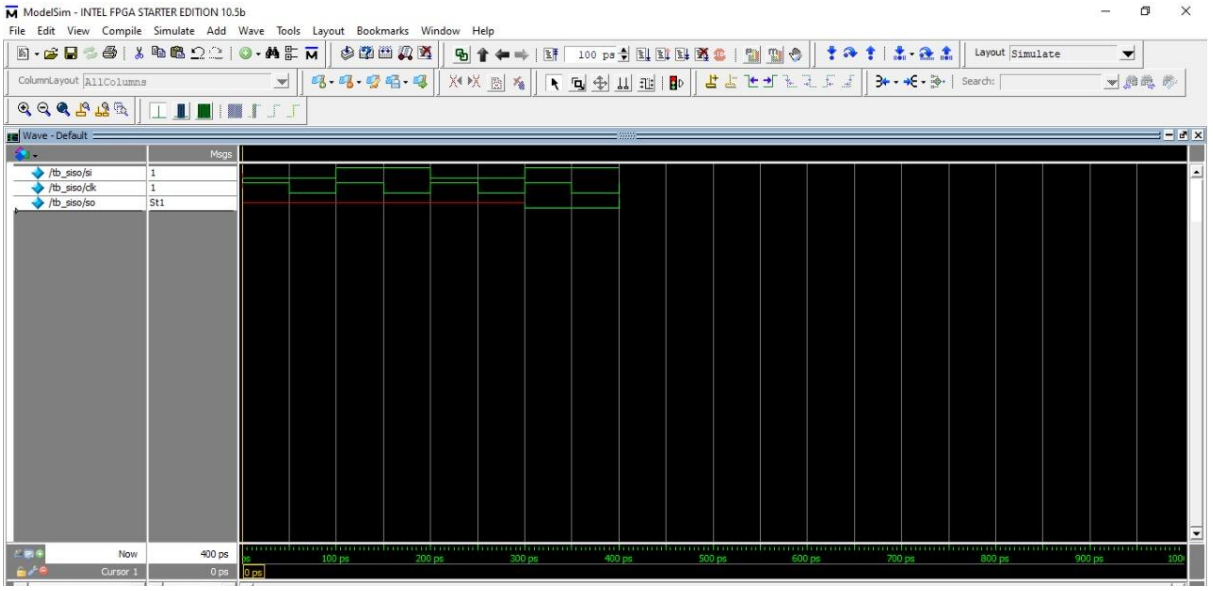
```
module tb_siso;
reg si, clk;
wire so;
```

```
siso s1(si,so,clk);
always #50 clk=~clk;
initial
begin
clk=1'b1;
        si=0;
#100    si=1;
#100    si=0;
#100    si=1;
end
endmodule
```

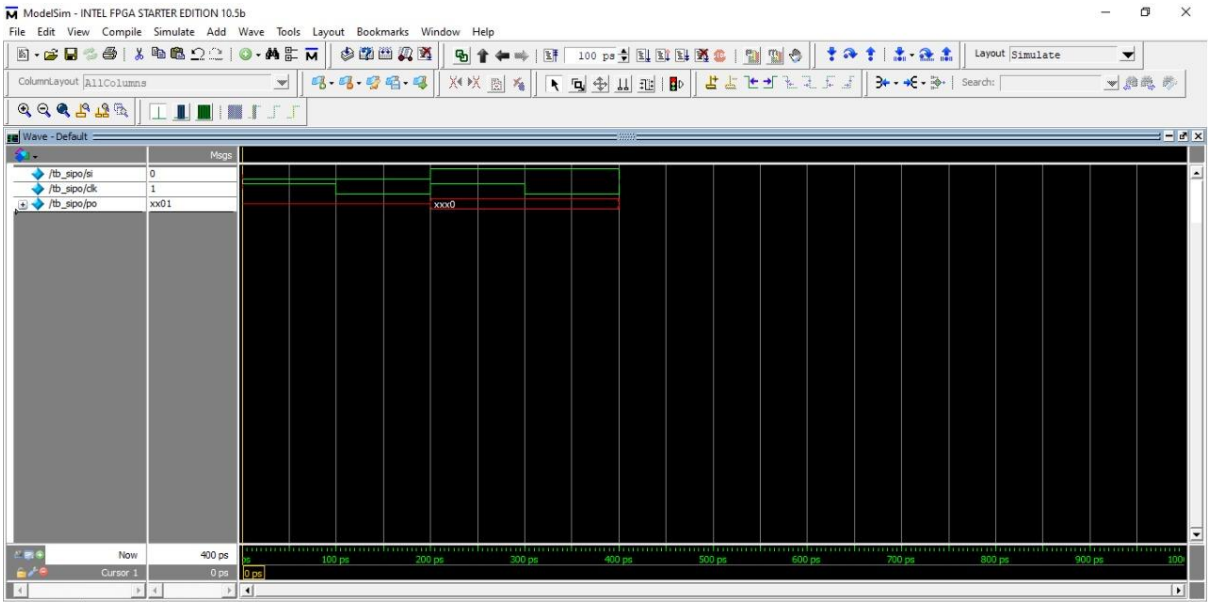


Output:

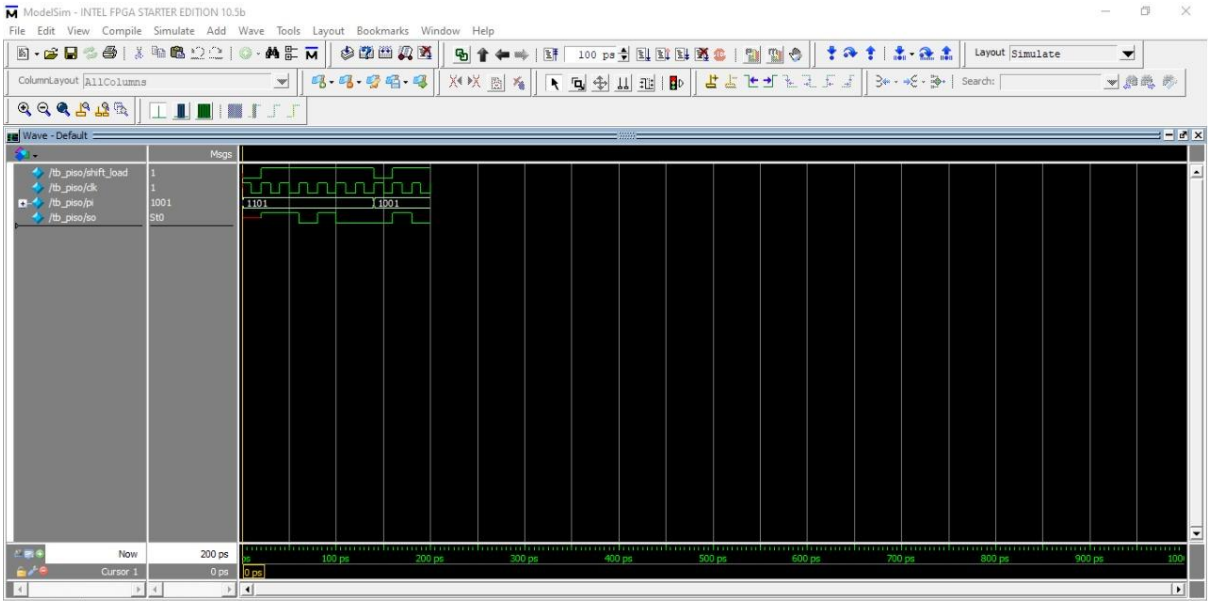
SISO:



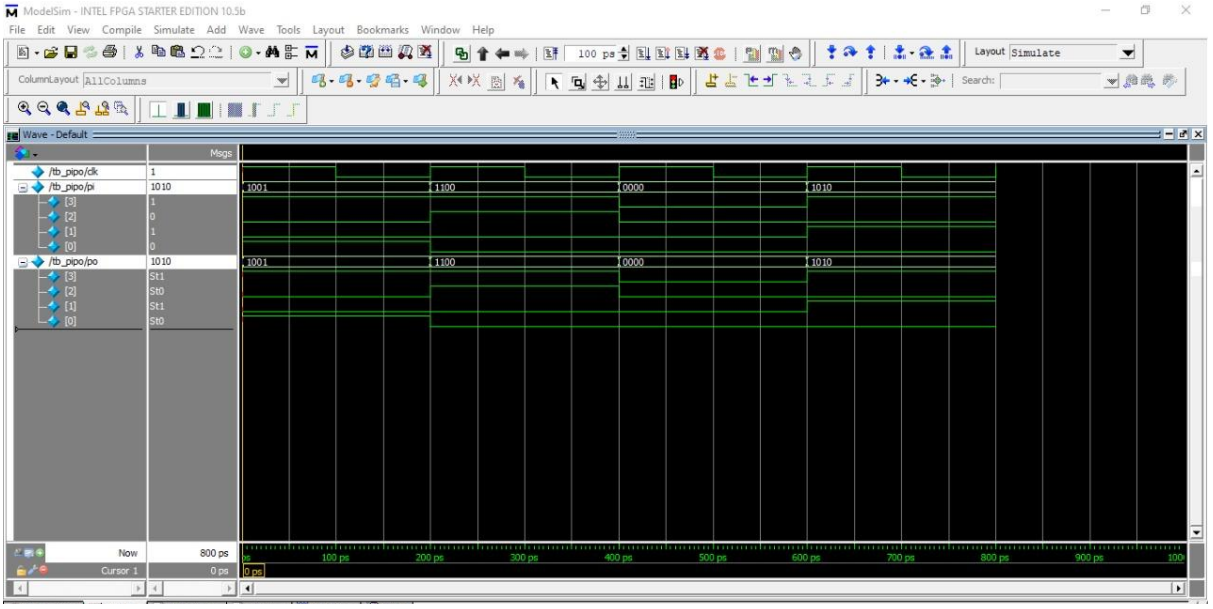
SIPO:


ModelSim - INTEL FPGA STARTER EDITION 10.5b
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
Wave - Default
Msgs
/tb_sipo/si
/tb_sipo/clk
/tb_sipo/po
xx01
xxx0
Now
400 ps
Cursor 1
0 ps

PISO:


ModelSim - INTEL FPGA STARTER EDITION 10.5b
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
Wave - Default
Msgs
/tb_piso/shift_load
/tb_piso/clk
/tb_piso/pi
/tb_piso/so
1001
St0
Now
200 ps
Cursor 1
0 ps

PIPO:



Result/ Inference:

From the given experiment, the shift register is verified using verilog programming.